

❏ OSI Model – Detailed Theory

The **OSI (Open Systems Interconnection) model** is a **conceptual framework** that standardizes the functions of a telecommunication or computing system into **seven distinct layers**. It helps network engineers and developers understand how data flows from one system to another.

❏ Overview of the OSI Layers

Layer	Name	Function	Example Protocols/Devices
7	Application	User interface and application services	HTTP, FTP, SMTP
6	Presentation	Data translation, encryption, compression	SSL/TLS, JPEG, MPEG
5	Session	Session management between applications	NetBIOS, RPC
4	Transport	End-to-end connections, reliability	TCP, UDP
3	Network	Routing, addressing	IP, ICMP, Routers
2	Data Link	MAC addressing, error detection	Ethernet, ARP, Switches, Bridges
1	Physical	Transmission of bits over media	Cables, Hubs, NIC, Fiber, Ethernet

❏ Layer 1: Physical Layer

❏ Responsibilities:

- Transmits raw bits (0s and 1s) over the communication medium.
- Deals with electrical, optical, and mechanical characteristics.
- Concerned with voltage levels, pin layout, and timing.

❏ Devices:

- Network Interface Cards (NICs)
- Hubs

- Cables (Ethernet, coaxial, fiber optics)
- Repeaters

☒ Standards:

- IEEE 802.3 (Ethernet)
- ITU V-series for modems

☒ Layer 2: Data Link Layer

☒ Responsibilities:

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Ensures error-free frame delivery over the physical layer.

- Performs MAC addressing and framing.
- Divided into **MAC (Media Access Control)** and **LLC (Logical Link Control)** sublayers.

☒ **Functions:**

- Error detection (using CRC)
- Flow control
- Frame synchronization

☒ **Devices:**

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Switches

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Bridges

☒ Protocols:

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Ethernet

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ARP

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PPP

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HDLC

☒ Layer 3: Network Layer

☒ Responsibilities:

- Handles logical addressing (IP addressing).
- Performs routing between different networks.
- Fragmentation and reassembly of packets.

☒ Functions:

- Packet forwarding and routing
- Logical addressing (IP)
- Path selection

☒ Devices:

- Routers
- Layer 3 Switches

☒ Protocols:

- IP (IPv4/IPv6)
 - ICMP
 - RIP, OSPF, BGP
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☒ Layer 4: Transport Layer

☒ Responsibilities:

- Provides reliable or unreliable delivery.
- Manages end-to-end communication and error recovery.

☒ Functions:

- Segmentation and reassembly
- Flow control (TCP windowing)
- Error control (acknowledgment and retransmission)

☒ Protocols:

- TCP (reliable)
 - UDP (unreliable)
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☒ Layer 5: Session Layer

☒ Responsibilities:

- Establishes, maintains, and terminates sessions between applications.
- Manages dialog control (half-duplex/full-duplex communication).

☒ Protocols:

- NetBIOS
 - RPC
 - PPTP
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⌘ Layer 6: Presentation Layer

⌘ Responsibilities:

- Translates data between application and network formats.
- Handles data encryption/decryption and compression.

☒ Functions:

- Syntax and semantics translation (e.g., EBCDIC to ASCII)
 - Encryption (e.g., TLS/SSL)
 - Compression (e.g., JPEG, MPEG)
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☒ Layer 7: Application Layer

☒ Responsibilities:

- Closest to the end user.
- Provides services for email, file transfers, web browsing, etc.

☒ Protocols:

- HTTP, HTTPS
 - FTP, SFTP
 - DNS, SMTP, POP3, IMAP
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☒ Mnemonic for Remembering the OSI Layers:

- **Top to Bottom:**
All People Seem To Need Data Processing
(Application, Presentation, Session, Transport, Network, Data Link, Physical)
- **Bottom to Top:**
Please Do Not Throw Sausage Pizza Away

(Physical, Data Link, Network, Transport, Session, Presentation, Application)

☒ Why the OSI Model Matters

- **Standardization:** Provides a universal language for network communication.
- **Troubleshooting:** Helps isolate network issues layer-by-layer.
- **Modular Engineering:** Enables modular design for software/hardware.
- **Interoperability:** Vendors can create interoperable systems across different platforms.